
OenoBench: A Wine-Domain Benchmark for Knowledge-Grounded Evaluation of Large Language Models

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Abstract

We introduce **OenoBench**, a knowledge benchmark for the wine domain comprising **3,266 multiple-choice questions** across six domain pillars (regions, grape varieties, viticulture, winemaking, producers, business) and four difficulty tiers. The benchmark is built from **38,104 atomic, source-anchored facts** extracted by 35 provenance-verified scrapers from authoritative sources spanning government registries (INAO, TTB, OIV), peer-reviewed journals, university research groups, and Wikipedia/Wikidata. Our central methodological contribution is a *LLM-driven, end-to-end pipeline* in which large language models reformat verified facts into questions and audit the result, but never serve as the source of truth: every claim traces to a URL with a tier-of-authority label, every question is generated by one of five complementary strategies (fact-to-question, template, comparative, scenario, distractor mining) distributed across five generator models (Claude, GPT, Gemini, Llama, Qwen + deterministic templates), and every question is scored by a nine-agent automated audit organised into four teams and calibrated against a human gold sheet via Cohen’s κ . We evaluate sixteen frontier configurations and report five findings: (i) overall accuracy ranges from 53% to 84%, with reasoning-mode **o3** on top at **83.6%**; (ii) reasoning-mode lift is concentrated in **DeepSeek R1 (+6.8 pp)** and largely absent for Claude Opus and Gemini Pro; (iii) the Anthropic family shows a **+9 pp self-preference** effect on its own questions while Google models show a **−8 pp inverse** effect; (iv) frontier open-weight models occupy the cost-vs-accuracy Pareto frontier together with proprietary reasoning models; and (v) every config gains $\approx$ **33 pp** on questions that the audit’s closed-book panel solves from world knowledge alone, revealing a parametric-recall ceiling that only the contextual, source-grounded slice avoids. We release the corpus, the audit findings, the human-review web app, and the construction code under CC-BY-SA-4.0.

1 Introduction

Large language models are increasingly evaluated on knowledge-intensive benchmarks, but the most widely-used resources — MMLU [6], GPQA [17], and their successors [20, 22, 16] — share three properties that limit their usefulness as the field matures: (i) they aggregate breadth at the expense of *depth* within any single discipline, (ii) they are typically authored or curated by a small team using a single generation pipeline, exposing them to systematic blind-spots and stylistic regularities that frontier models can learn to exploit [15, 21], and (iii) the questions are increasingly suspected of contamination through web-scale pre-training corpora [18, 13, 3]. As models grow more capable, the marginal value of a benchmark depends increasingly on *how* it was constructed, not only on what it contains.

We argue that domain-specialised benchmarks built around bodies of knowledge that are *externally validated by humans* — through certification, regulation, or peer-reviewed practice — offer a useful complement to broad knowledge benchmarks. Examples already exist in medicine [9, 10], law [5], and finance [8]. We extend this line of work to a domain that has so far received little attention from the benchmarking community, despite being unusually well-defined: **wine**.

Why wine. Wine is a multi-disciplinary subject that integrates plant biology, geology and climate science, organic chemistry, food microbiology, sensory neuroscience, regulatory law, and business economics. It is one of very few domains for which (a) authoritative governmental and inter-governmental sources publish structured taxonomic data (e.g. INAO, OIV, TTB, regional consortia), (b) university research groups (notably UC Davis, Geisenheim, Bordeaux Sciences Agro) produce open scholarly literature, and (c) professional certification bodies define a graded competence ladder (WSET Levels 1–4, Court of Master Sommeliers, Institute of Masters of Wine) whose published syllabi can be used to ground what “expert” should mean. Together these properties make it possible to build a benchmark in which every question is traceable to a verifiable source, while still spanning the recall–reasoning–application spectrum that distinguishes a strong taster, winemaker, or buyer from a memorised glossary.

Contributions.

1. **A 3,266-question wine benchmark** covering six pillars (regions, varieties, viticulture, winemaking, producers, business), built from 38,104 atomic facts whose provenance is recorded down to the source URL and tier of authority (Section 3).
2. **An LLM-driven, fact-grounded question-generation pipeline** that combines five generation strategies and five generator models. The LLM’s role is reformatting verified facts, *not* providing them; an explicit closed-book gate flags questions solvable from world knowledge alone, and a per-strategy quota prevents pipeline mono-culture.
3. **A nine-agent automated quality audit** (Section 4) consisting of static, statistical, and tri-judge agents organised into four teams, calibrated against a human gold sheet using Cohen’s κ . The audit removed 341 questions and re-labelled the difficulty of 1,259 in the final release. Crucially, the audit also revealed that one of its own agents (closed-book solvability) over-reports leakage relative to humans; we kept the affected questions with explicit disclosure rather than dropping them, and demonstrate the consequences in the evaluation.
4. **A bias-aware evaluation across sixteen frontier configurations** (Section 5) reporting overall accuracy, reasoning-mode lift, a Self-Preference Score that quantifies own-vs-other-generator gaps, a cost-efficiency Pareto analysis, and a closed-book vs. source-grounded contrast that isolates parametric memorisation from contextual reasoning.

OenoBench, the construction and audit pipelines, the multi-agent audit findings, and the human-review web application are released under CC-BY-SA-4.0 at <https://huggingface.co/datasets/oenobench/oenobench> with full source at <https://github.com/nikitahudov/oenobench>.¹

2 Related Work

Broad knowledge benchmarks. MMLU [6], BIG-Bench [20], AGIEval [22], GPQA [17], HELM [12], and Humanity’s Last Exam [16] measure factual breadth and reasoning across many subjects. As headline scores have saturated, attention has shifted toward harder reasoning benchmarks and toward auditing the construction process. Our work is complementary: rather than maximising difficulty across many disciplines, we maximise depth within a single externally-validated knowledge domain, and treat *construction-time bias control* as a first-class methodological contribution rather than a downstream decontamination step.

Domain-specific benchmarks. MedQA [9], PubMedQA [10], LegalBench [5], and FinanceBench [8] demonstrate the value of grounding evaluation in a regulated profession with clear sources of truth. We adopt their organising principle — expert syllabus as taxonomy, source-traceable facts as backbone — and add multi-model question generation, an automated multi-agent audit, and

¹For double-blind submission both URLs are mirrored to anonymous fora; see Appendix E for the camera-ready swap.

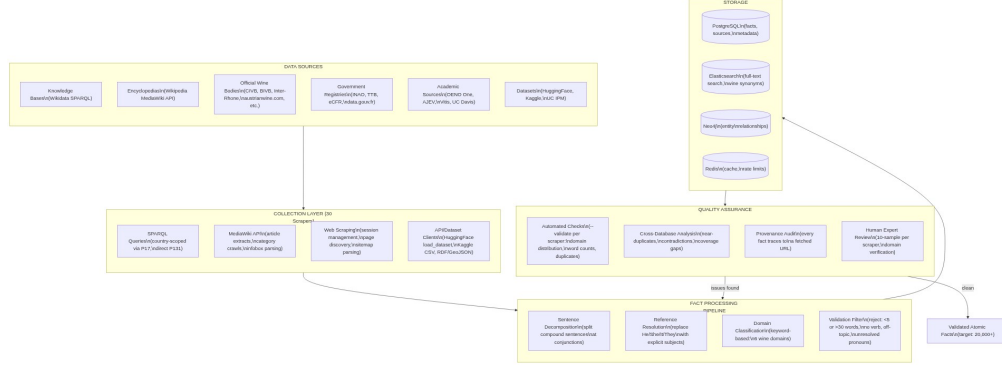


Figure 1: End-to-end pipeline. Authoritative sources (top) are scraped by 35 source-specific extractors that share an atomic-fact decomposition and validation pass. Generated questions are gated by a closed-book pre-screen and a nine-agent audit before being tagged for release.

self-preference reporting as additional bias controls. To our knowledge, OenoBench is the first such benchmark for the wine domain.

LLM-as-judge bias and self-preference. A growing literature documents that models tend to favour outputs stylistically similar to their own [15, 21, 19]. This biases benchmarks where the same model generates and grades. We mitigate this in two ways: by distributing generation across five models (Claude, GPT, Gemini, Llama, Qwen) plus deterministic templates, and by reporting per-model self-vs-other accuracy gaps as a first-class diagnostic (Section 5.3).

Benchmark contamination and solvability. Pre-training exposure inflates apparent capability [18, 13, 3]. We mitigate contamination in two ways: (i) generating questions from atomic facts rather than copying source text (a fact-echo agent flags any question whose longest common substring with the source exceeds 65%), and (ii) running an explicit *closed-book solvability* pre-screen and audit, allowing us to report a separate per-config accuracy on world-knowledge-solvable items (Section 5.5). We further show that the same audit, when calibrated against a human gold sheet, reveals that LLM judges themselves over-report leakage by an order of magnitude — a finding with broader implications for the LLM-as-judge paradigm.

Dataset documentation. We follow Datasheets for Datasets [4], ship a Croissant manifest [1] with the HuggingFace release, and publish a full audit findings table to support inspection in the spirit of Model Cards [14]. Wine specifically has appeared in NLP only as a substrate for review-text generation, recommendation [2], taxonomy learning [11], and judge-reliability studies [7]; OenoBench is the first benchmark that targets structured factual knowledge of viticulture, winemaking, and wine regions evaluated against frontier LLMs.

3 Dataset Construction

OenoBench is constructed by an LLM-driven pipeline (Figure 1) in two stages: (1) a *fact-collection* stage that builds a corpus of atomic, provenance-tagged statements from authoritative sources, and (2) a *question-generation* stage that converts those facts into benchmark questions using five complementary strategies and five generator models. The pipeline is designed around a single methodological commitment: large language models are used as *rephrasers* and *auditors*, never as the source of truth. Every fact in the corpus traces to a URL with a tier-of-authority label, every question is anchored in one or more such facts, and every question is scored by a nine-agent automated audit (Section 4) before release.

3.1 Domain taxonomy

We organise wine knowledge into six *domain pillars*, chosen to align with the WSET Level 3 and Diploma syllabi: (i) **wine regions** (appellations, classifications, geography, climate, soils); (ii) **grape**

Table 1: Atomic-fact corpus: 38,104 facts from 35 scrapers, six domain pillars, three source tiers.

Domain pillar	Facts	Share	Tier 1	Tier 2/3
Wine regions	18,943	49.7%	23.4%	76.6%
Producers	6,215	16.3%	9.1%	90.9%
Grape varieties	5,959	15.6%	18.8%	81.2%
Viticulture	3,635	9.5%	12.0%	88.0%
Wine business	1,985	5.2%	31.4%	68.6%
Winemaking	1,367	3.6%	16.7%	83.3%
Total	38,104	100.0%	19.6%	80.4%



Figure 2: Atomic-fact extraction pipeline. Inputs are heterogeneous (HTML pages, RDF graphs, SPARQL responses, CSV dumps); outputs are uniform single-assertion sentences with entity tags, domain label, source URL and tier.

varieties (ampelography, parentage, regional distribution, sensory profile); (iii) **viticulture** (vineyard practice, training systems, pests and diseases, vintage effects); (iv) **winemaking** (vinification choices, fermentation, ageing, faults); (v) **producers** (estates, négociants, co-operatives, key houses); and (vi) **wine business** (markets, regulation, distribution, economics). Each domain has a separate target share informed by syllabus weighting and source availability; the realised distribution is reported in Table 1.

3.2 Source tiering and provenance integrity

Every fact is tagged with one of three source tiers, mirroring evidence-based-medicine practice:

- **Tier 1 (official, 19.6%)**: government registries and inter-governmental bodies (INAO, OIV, TTB, regional consortia, the UC Davis AVA database).
- **Tier 2 (authoritative, 76.6%)**: Wikipedia (34.3% of facts), Wikidata SPARQL (31.0%), wine-body and consortium websites, peer-reviewed journals (*OENO One*, *Vitis*, *Australian Journal of Grape and Wine Research*).
- **Tier 3 (reliable, 3.8%)**: curated open datasets on HuggingFace and Kaggle, secondary reference databases.

A 2026-04 internal audit identified 19 of an earlier 35-scraper set whose “scraped” facts were in fact hard-coded LLM-generated content. All 19 were rebuilt against live sources, the corpus was purged of 24,563 $\rightarrow$ 16,702 facts and rebuilt to the current 38,104, and the entire fact corpus reported here passes a per-source URL trace check. We document this rebuild because we believe that **provenance auditing should be a first-class step in any LLM-built benchmark**: model-generated content masquerading as scraped data is a contamination vector that conventional dataset documentation does not detect, and the fix is mechanical only if scraper code is released alongside the data.

3.3 Atomic fact extraction

Source pages are decomposed into atomic facts by a five-stage pipeline (Figure 2): (1) sentence splitting at conjunctions and clause boundaries, (2) reference resolution (pronouns and demonstratives are replaced with their entity referents from article context), (3) domain classification into one of the six pillars, (4) a length-and-predicate validator that rejects facts shorter than five words, longer than fifty words, missing a verb, or carrying dangling references, and (5) an on-topic filter parameterised by region-specific keyword sets to prevent cross-contamination (e.g. Austrian content appearing in a Bordeaux scraper). Wikidata SPARQL queries use direct country relations (P17) rather than transitive administrative parents (P131*); the latter caused severe cross-region leakage in early scrapers and is documented as a cautionary case in Appendix B.

Facts are required to be (a) *atomic*, one assertion per sentence; (b) *entity-tagged*, with subject and object linked to canonical IDs in the wine knowledge graph; and (c) *source-faithful*, a paraphrase of

the original, never a verbatim copy. This last constraint is enforced both at extraction time and at audit time (Section 4, agent A3).

3.4 Question generation: five strategies, five models

A single-strategy, single-model benchmark inherits the blind-spots of its pipeline. We mitigate this by combining *five strategies* and *five generator models*, with per-strategy and per-model quotas enforced by an orchestrator that resumes safely across runs. Strategies are:

- **Fact-to-question (58.4%, 1,909 questions):** the LLM rewrites a single verified fact as a multiple-choice question; preserves grounding but tends toward recall.
- **Distractor mining (12.4%, 405):** confusable-entity sampling produces wrong answers that share the right answer’s category and dimension, raising plausibility and forcing fine-grained discrimination.
- **Template (11.9%, 389):** forty-five deterministic parameterised templates across the six pillars; pure entity substitution, zero LLM creativity — a baseline against which neural generation is measured.
- **Scenario synthesis (9.8%, 319):** coherent fact clusters are converted into applied multi-fact reasoning prompts (e.g. a sommelier service decision, a winemaker blending decision, a viticulturist vintage call). Domain-specific scenario types prevent persona-content mismatch.
- **Comparative (7.5%, 244):** entity-affinity scoring pairs related entities (e.g. two Burgundy premiers crus, two Rhône grapes) and asks “which differs in X” or “what do both share”.

The five generator models are Claude Opus 4.7, GPT-5, Gemini 2.5 Pro, Llama 3.1 405B, and Qwen 3.5 235B, each accessed through a unified OpenRouter client with consistent temperature and top-*p*. Generator distribution across the corpus is approximately Qwen 20%, Llama 19%, Claude 19%, ChatGPT 17%, Gemini 13%, deterministic templates 12%.

Closed-book gate. Questions targeting difficulty levels 1 and 2 are passed through a *closed-book pre-screen* before admission: a separate LLM (Claude Sonnet for L1, Opus for L2) attempts the question with no source fact in context, and questions it answers correctly are tagged `closed_book_solvable` and reserved into a quota (target <25%). Difficulty levels 3 and 4 skip the gate by protocol, since high-difficulty items are expected to be answerable only with the supporting fact. The audit’s B2 agent (Section 4) re-runs this check at corpus scale on a panel of three judges; we discuss its calibration against humans and the implications for the eval contrast in Section 5.5.

3.5 Final corpus: release_v1.2

After generation, audit, and post-eval review, the released corpus `release_v1.2` contains **3,266 questions**. The distribution along three axes is:

Table 2: Composition of the released corpus.

Domain	N	%	Difficulty	N	%
Wine regions	1,108	33.9%	L1 (entry)	694	21.2%
Grape varieties	766	23.5%	L2 (intermediate)	894	27.4%
Producers	515	15.8%	L3 (advanced)	678	20.8%
Viticulture	502	15.4%	L4 (expert)	1,001	30.6%
Wine business	250	7.7%			
Winemaking	187	5.7%			
Total	3,266	100%	Total	3,266	100%

The corpus underwent two post-generation passes: a 9-agent automated audit (Section 4) that dropped 341 questions and re-labelled the difficulty of 1,259, and a post-evaluation review of 97 items that were answered incorrectly by all 16 evaluation configurations (Section 5, Appendix C), of which 54 were removed as defects, 9 were dropped after a borderline-review pass, and the remainder retained as legitimately hard items.

4 Multi-Agent Quality Audit

The audit pipeline runs every generated question through nine automated agents organised into four teams (Figure 3). Each agent emits a per-question signal in $\{\text{PASS}, \text{WARN}, \text{FAIL}\}$ and signals are calibrated against a human-reviewed gold sheet using Cohen’s κ . Agents whose human agreement falls below threshold are downweighted to advisory-only, on the principle that an automated judge whose verdict humans cannot reproduce should not gate question admission. A separate human-review web application (Appendix D) collects the gold-sheet ratings from WSET-Diploma reviewers and is itself part of the released artifact set.

4.1 Team architecture

Table 3: Nine audit agents across four teams. Static agents use no LLM; tri-judge agents use a panel of three frontier models with majority voting.

Team	Agent	Function
A. Static	A1 LexicalHygiene	Regex sweep for vague phrasing, marketing language, meta-questions.
	A2 BiasStats	χ^2 on correct-answer position; Mann–Whitney U on correct-vs-distractor length.
	A3 FactEcho	Longest-common-substring ratio between question text and source fact (FAIL at $\text{LCS} \geq 0.65$).
	A4 TemplateFingerprint	POS-bigram logistic regression detector for template-induced stylistic regularities (held-out AUC 0.84).
B. Tri-judge	B1 TriJudgeAnswer	Claude / GPT / Gemini panel reads question + source fact and votes on the marked-correct option.
	B2 ClosedBookSolvability	Same panel, source fact <i>withheld</i> : can the question be answered from world knowledge alone?
C. Determ.	C2 CategoryLeak	Wine-type distractor validation; flags red/white/sparkling category mismatches.
	C4 DifficultyAudit	Gemini Pro re-rates difficulty; FAIL/WARN when delta ≥ 2 from assigned.
D. Population	D1/D3 Skew & SPS	Country/sub-domain over-representation; per-generator self-vs-other accuracy.
	B3 UbiquityRisk	Internationally-grown grape stem $\times$ region-class answer $\Rightarrow$ ambiguity flag.

Five additional agents (C1 DistractorDifficulty, C3 SourceSwap, B4 Ambiguity, D2 DedupCalibration, plus an escalation-only B5 verifier) are implemented but *escalation-gated*: they are invoked only when an upstream agent flags a corpus-level concern. This deferred-execution policy keeps routine audit cost bounded ($\sim$ \$76 for the full corpus audit reported here, 5h 22m wall) while preserving full coverage when a defect class is suspected. The full audit code, prompts and thresholds are released alongside the corpus (Appendix B).

4.2 Gold-sheet calibration

A human gold sheet of 50 stratified questions per release is rated by a WSET-Diploma reviewer against eight rubrics: `answer_correct`, `distractors_plausible`, `not_ambiguous`, `source_faithful`, `needs_source`, `no_vague_language`, `labels_correct`, and `verbatim_copy`. For each automated agent we compute Cohen’s κ against the corresponding human rubric on the gold sheet. **Agent signals with $\kappa < 0.6$ are downweighted to advisory-only and excluded from gating decisions.**

The most consequential calibration result concerns agent **B2 ClosedBookSolvability**: against the human reviewer, who solved only $\approx 12\%$ of the gold-sheet questions without seeing the source fact, the LLM tri-judge panel reported $\approx 83\%$ solvability, yielding $\kappa \approx 0.007$. We interpret this not as a defect of the benchmark but as a property of the *evaluator*: frontier LLM judges have absorbed enough wine knowledge during pre-training to over-attribute “world knowledge” to questions that humans find non-trivial. We therefore retain the 1,601 B2-flagged questions in the released corpus, document the finding in the datasheet, and report a closed-book vs. source-grounded contrast in Section 5.5 that turns the audit’s blind spot into a calibrated diagnostic of memorisation reliance.

4.3 Release-cycle results

The 9-agent audit on `release_v1.1` (3,670 candidate questions, run 2ba38269) produced the verdicts in Table 4.

Table 4: Audit findings on `release_v1.1` and the resulting drop / re-label policy.

Audit signal	Triggered FAIL	Action
A1 LexicalHygiene (vague phrasing)	60	drop
A3 FactEcho (verbatim copy)	63	drop
B1 TriJudgeAnswer (wrong answer key)	47	drop
C2 CategoryLeak (wine-category leak)	9	drop
B3 UbiquityRisk (grape $\times$ region ambiguity)	183	drop
Distinct dropped (with overlap)	341	
C4 DifficultyAudit (relabel from 1,252 LLM hits + 7 human)	1,259	re-label
B2 ClosedBookSolvability (kept w/ disclosure)	1,601	disclose

After drops, a follow-up post-evaluation pass (Section 5 and Appendix C) audited the 97 questions that all 16 evaluation configs answered incorrectly: 54 were removed as defects (wrong ground truth, all-correct options, duplicate options), 9 were dropped on borderline-review and the remaining 34 retained as legitimately hard. The final released corpus is 3,266 questions.

The C4-driven difficulty re-label was decisive: it shifted the corpus from 14% L3+L4 items (the original generator-assigned distribution) to **51% L3+L4** (Table 5), bringing the hardest-tier counts above the entry-tier counts and substantially sharpening difficulty-stratified analyses (Section 5.1).

Table 5: Difficulty distribution before and after C4 re-label.

	Generator-assigned	Post-relabel	Δ
L1 (entry)	1,261 (38%)	694 (21%)	−567
L2 (intermediate)	1,559 (47%)	894 (27%)	−665
L3 (advanced)	218 (7%)	678 (21%)	+460
L4 (expert)	291 (9%)	1,001 (31%)	+710

4.4 Calibration philosophy

We emphasise the design choice underlying these numbers. A naive “automated audit” interpretation would have dropped all 1,601 B2-flagged questions, which would have removed nearly half the corpus and biased the remainder toward whatever idiosyncratic items the LLM panel happened not to know. By calibrating B2 against humans and treating it as an evaluator-side signal, we preserve dataset value while giving downstream users a principled way to interpret leakage. The audit’s role, on this reading, is not to be the final arbiter of quality, but to surface signals *that humans validate* as quality-relevant, and to disclose clearly the signals where humans and LLMs disagree.

5 Evaluation

We evaluate **16 frontier configurations** on the 3,266-question release in a single end-to-end run (run id 8b0a0864-f3c6-4ec5-8f3d-e30271b8c3a0, 120m 34s wall, 52,256 LLM calls, total cost \$98.33 effective). The slate (Slot 1–16) combines six within-family cost pairs (e.g. Claude Opus vs. Haiku, Gemini 2.5 Pro vs. Flash, GPT-5 vs. GPT-5-mini, Llama 3.3 70B vs. 3.1 8B, Qwen 2.5 72B vs. 7B), four reasoning-mode configurations (o3, deepseek-r1, gemini-2.5-pro-thinking, claude-opus-4.7-thinking), and two additional standard models (deepseek-v3, mistral-large-2411). Every config sees the same 3,266 questions, with single-letter output (A–D) at `max_tokens=5` and a five-stop fallback. We organise the evaluation around five questions, one per subsection.

5.1 Overall capability ranking

OenoBench cleanly stratifies the slate (Figure 4). The top tier is led by **o3** at **83.6%**, followed by GPT-5 (82.8%), gemini-2.5-pro-thinking (82.6%), Gemini 2.5 Pro standard (81.7%), and Claude Opus 4.7 (81.0%, tied between standard and thinking modes). A clear band separates frontier ($\geq 80\%$) from mid-tier (mid-60s to high-70s: Mistral Large 69.1%, Llama 70B 67.1%, Qwen 72B 67.4%, DeepSeek V3 70.3%, Gemini Flash 75.1%, GPT-5-mini 78.4%) and small-tier (low-50s to low-60s: Llama 8B 60.5%, Qwen 7B 57.0%, Claude Haiku 53.3%). No config performs at random; the spread between best and worst is 30 pp, comparable to the spread observed on GPQA [17] but on a domain corpus $30\times$ smaller and fully provenance-tagged.

The Haiku 4.5 result deserves a caveat: parse rate dropped to 80.3% (vs. 96–100% for every other config), driven by truncated outputs at `max_tokens=5`. We treat unparseable responses as wrong; recovering the 19.7% lost responses would lift Haiku into the mid-60s. Difficulty-stratified accuracies (Appendix C, Table 9) confirm clean stratification on re-labelled difficulty: from 93.6% (L1) down to **58.7% on L4 expert items**, with frontier configs holding 68–71% on L4 and small configs at 39–45%.

5.2 Reasoning-mode lift

Reasoning-mode lift is concentrated in one family (Figure 5). **DeepSeek R1 vs. V3**: +6.8 pp (95% CI [+4.6, +8.8]), the only paired contrast whose CI excludes zero. Each of **o3 vs. GPT-5** (+0.8 pp [-1.0, +2.6]), **Gemini 2.5 Pro thinking vs. standard** (+0.9 pp [-1.0, +2.8]), and **Claude Opus 4.7 thinking vs. standard** (−0.1 pp [-1.9, +1.8]) is statistically indistinguishable from zero. We attribute the DeepSeek result to V3 leaving substantial headroom on this domain (70.3%) which R1’s chain-of-thought can recover; the three frontier cases are likely already near the corpus ceiling on parametrically-recallable items. Per-difficulty breakdowns (Appendix C, Table 10) confirm this: reasoning helps DeepSeek across all four difficulty tiers, but the frontier reasoning configs are flat against their standard counterparts at L1–L3 and gain at most 1–2 pp at L4. The cost-vs-lift trade-off is unfavourable for reasoning configs in this domain except DeepSeek (Section 5.4).

5.3 Self-preference bias

We compute the *Self-Preference Score (SPS)* per config m as $SPS(m) = \text{Acc}(m \mid Q_{\text{own}}) - \text{Acc}(m \mid Q_{\text{other}})$ where Q_{own} collects questions generated by m ’s family and Q_{other} its complement. A pipeline that successfully neutralises generator fingerprints should produce $|SPS|$ near zero.

The result (Figure 6) is family-dependent. **Anthropic** models show a strong positive SPS: +9.1 pp [+6.1, +12.1] for Opus standard, +9.7 pp [+6.5, +13.1] for Opus thinking, +10.0 pp [+5.7, +14.5] for Haiku. **OpenAI** is statistically indistinguishable from zero across o3, GPT-5, and GPT-5-mini (−1.7 to +1.7 pp). **Google** models show an unexpected *negative* SPS: −6.1 pp [−10.7, −1.8] for Gemini Pro, −8.0 pp [−12.3, −3.8] for Gemini Pro thinking, and −10.4 pp [−15.3, −5.6] for Gemini Flash. The cross-family matrix (Figure 6, left) traces the cause: Google-generated questions are uniformly harder for everyone (46–76% across rows), while Anthropic-generated questions are uniformly easier for Anthropic (77.4%) but typical for others (65–86%). We interpret the Anthropic +9 pp gap as a residual stylistic fingerprint that pre-release paraphrase passes did not neutralise; the Google −8 pp gap suggests Google generators produce questions that are harder *by phrasing*, not by leak. Both findings are first-class disclosures with the released corpus.

5.4 Cost efficiency

The 16-config slate spans nearly four orders of magnitude in evaluation cost: llama-3.1-8b costs \$0.01 for the full run while gemini-2.5-pro costs \$29.47 (Figure 7). The Pareto frontier consists of five configs: **Llama 3.1 8B** (60.5% / \$0.01), **Gemini 2.5 Flash** (75.1% / \$0.12), **GPT-5-mini** (78.4% / \$2.82), **Claude Opus 4.7** (81.0% / \$3.35), and **o3** (83.6% / \$11.80). Two of the most expensive configs (Gemini 2.5 Pro standard at \$29.47 and GPT-5 at \$21.90) are dominated by cheaper Pareto neighbours — Gemini Pro by o3, and GPT-5 by Opus / o3 / GPT-5-mini — because their additional spend buys output tokens, not better wine answers. Reasoning-mode lift (Section 5.2) translates into a Pareto win only for DeepSeek R1, which is itself dominated by GPT-5-mini in this corpus. The overall efficient-accuracy boundary therefore favours *small proprietary* (GPT-5-mini,

Opus standard) and one reasoning model (o3) over either reasoning-everywhere or scale-everywhere strategies.

5.5 Closed-book vs. source-grounded performance

Section 4.2 described the calibration finding that the audit’s B2 ClosedBookSolvability agent reports leakage at $\sim 83\%$ while human reviewers solve only $\sim 12\%$ of the same items closed-book. That gap is itself a measurement: it tells us what a panel of frontier LLMs already *know* about wine, separately from what the questions test. We exploit this by partitioning the corpus into the 1,601 B2-flagged questions (*closed-book solvable* per the LLM panel) and the 1,665 contextual / source-grounded questions, and reporting per-config accuracy on each slice.

The contrast (Figure 8) is the largest single effect we observe in the entire evaluation. The mean per-config δ (closed-book minus contextual) is **+32.6 pp**, ranging from +26.6 pp (GPT-5) to +39.6 pp (DeepSeek V3), with every CI excluding zero. Two interpretations are mutually consistent. *First*, the B2-flagged slice is meaningfully easier — not because all 1,601 questions are trivial, but because they overlap heavily with the parametric wine knowledge that frontier models acquired during pre-training. *Second*, the contextual slice is where wine knowledge actually has to be reasoned through the source fact: **Claude Opus and o3 score $\sim 70\%$ on the contextual slice, the small models drop to $\approx 45\%$** , and the inter-config spread widens from 27 pp on the easy slice to 33 pp on the hard slice. For benchmark consumers, the contextual slice is the more discriminative test of wine reasoning capability; for benchmark contributors, the B2-flagged slice is a calibrated indicator of how much of any given LLM’s apparent “wine knowledge” is parametric recall versus in-context reasoning.

We close the loop on the audit (Section 4) here: the methodological consequence of releasing both slices is that downstream users can choose to evaluate only on the contextual slice if their research question is reasoning-with-context, or on the union if it is total wine capability. The corpus supports both readings.

6 Applications and Future Work

OenoBench is a benchmark first, but the construction pipeline behind it is intended to be reusable. We outline three threads that build on the released artifacts and methodology, in approximate order of immediate feasibility.

6.1 Closing the wine-knowledge gap in low-cost models

The cost-vs-accuracy frontier (Section 5.4) shows that small open-weight models (Llama 8B, Qwen 7B, Llama 70B) trail proprietary frontier models by 20–30 pp on this benchmark, even though the gap shrinks to 10–15 pp on broad benchmarks like MMLU. The 38,104-fact corpus, with paragraph-length explanations grounded in source URLs, is a natural supervised-fine-tuning target for a *wine-specialist* small model. Our preliminary plan is a two-stage recipe: (i) continued pre-training on the atomic-fact corpus to inject named-entity coverage, and (ii) supervised fine-tuning on the held-out quartile of OenoBench questions plus their fact provenance. The contextual / source-grounded slice (Section 5.5) provides a clean evaluation surface that is harder to game by parametric memorisation, so the gain measured against frontier models is a faithful proxy for genuine wine reasoning. We expect a Llama-70B-class model to recover the bulk of the GPT-5-mini gap at $\approx 1\%$ of proprietary inference cost, opening realistic deployment paths for sommelier-tooling and wine-education applications that are cost-sensitive at high volume.

6.2 Case-based evaluation with industry partners

Multiple-choice factual tests have ceilings: even a perfect OenoBench score does not establish that a model can act usefully in real workflows. We are scoping an **applied case-based dataset** in collaboration with industry partners along three personas:

- *Sommelier* $\rightarrow$ *customer service*. Realistic consumer prompts (“what red would pair with a duck confit and please my mother who likes Pinot but doesn’t want anything tannic”), rated by certified sommeliers against the model’s recommendation, justification, and substitutions.

- *Winemaker* → *blending decisions*. Vintage write-ups, lab analyses, and regulatory constraints fed in as context; model outputs a blending plan; rated by working winemakers on technical correctness, regulatory compliance, and stylistic intent.
- *Buyer / merchant* → *vintage strategy*. Market data, weather records, and producer context; model outputs an allocation recommendation; rated by experienced buyers against actual market outcomes the following season.

Such a dataset cannot be built from scraped facts alone — it requires domain-expert annotation of free-form outputs. The construction pipeline of OenoBench (multi-strategy generation, multi-agent audit, gold-sheet calibration) carries over: scenario-style generation templates can produce candidate prompts; the same nine-agent audit can flag obvious failure modes; and the human-review web app (Appendix D) is already designed to support multi-reviewer IRR aggregation. We treat OenoBench as the *factual-recall floor* of an eventually two-tier evaluation: factual knowledge must be near-saturated before applied reasoning can be validly measured.

6.3 Generalising the LLM-driven scrape-to-audit pipeline

The most consequential contribution of this work is methodological, not domain-specific. The pipeline elements — (i) a tier-of-authority source taxonomy, (ii) atomic-fact extraction with paraphrase enforcement, (iii) multi-strategy multi-model question generation with a per-model quota, (iv) an automated multi-agent audit calibrated against humans via κ , and (v) a closed-book pre-screen that isolates parametric leakage — are domain-agnostic. We have implemented them as separable modules with clear interfaces (`src/scrapers/_fact_processing.py`, `src/generators/orchestrator.py`, `src/qa/orchestrator.py`, `src/review_app/`); replacing the source-tier configuration and the domain taxonomy is sufficient to retarget to a different field.

We see three near-term targets. First, *horticulture and agronomy* share many of wine’s structural features (regulated appellations, peer-reviewed agronomic journals, certification ladders for advisors) and would benefit from the same provenance discipline. Second, *regulated medicine* (clinical guidelines, drug formularies) has the strongest tier-of-authority structure in any field and the most acute need for source-traceable benchmarks; medical benchmarks today rely on board-exam questions which are not necessarily aligned with practice. Third, *financial regulation and accounting* has tiered authority (FASB, IFRS, jurisdictional filings) and a clear professional ladder (CFA, CPA), and would benefit from a benchmark whose questions are explicitly anchored in citable regulation. In each case, the pipeline reduces what is currently a months-long expert-authoring effort to a weeks-long curation effort, provided the tier-of-authority configuration is done carefully. The released code and the methodology in Sections 3 and 4 are intended as a recipe for that retargeting.

7 Limitations

Language and cultural scope. OenoBench is currently English-only. The Tier-2 source mix (Wikipedia 34%, Wikidata 31%) is itself English-skewed, and the corpus inherits a geographic bias toward European (especially French, Italian, Spanish) and New-World canonical regions. Emerging wine cultures (China, India, much of Eastern Europe) are under-represented in absolute counts. Multilingual extension is the largest single gap and is the most natural next-version target.

Source skew within domains. Producer and wine-business domains are smaller (515 and 250 questions respectively) than wine regions (1,108) and grape varieties (766) because authoritative producer-level and business-level open data is scarcer than appellation-level data. The C4 difficulty re-label (Section 4) shifted the corpus toward harder items but cannot create coverage that did not exist; the per-domain accuracy ceiling on *wine_business* (**56.5% mean across 16 configs**, vs. 79% on producers) reflects both the hardness of the domain and its sparser fact pool.

Snapshot vs. moving target. Wine regulation, classifications, and producer ownership change on a multi-year cadence (INAO appellation revisions, DOCG promotions, estate sales). The released corpus is a snapshot dated 2026-04-01 and will require periodic re-extraction; we release the scrapers and provenance metadata so this is mechanically reproducible.

Closed-book audit calibration. The B2 ClosedBookSolvability agent shows $\kappa \approx 0.007$ with humans on the gold sheet (Section 4.2). Rather than drop the affected questions, we keep them with explicit disclosure and report the closed-book vs. contextual contrast in Section 5.5. Downstream users who prefer a strictly source-grounded subset can use the contextual slice; those who want pure recall can use the closed-book slice. The documentation is a load-bearing part of the dataset, not a footnote.

Self-preference is not fully neutralised. The Anthropic +9 pp positive SPS (Section 5.3) implies the multi-model generation strategy did not fully neutralise stylistic fingerprints. We disclose per-config SPS as a first-class diagnostic rather than removing the affected items, but downstream comparisons between Anthropic and other-family models on the full corpus should be treated with the SPS in mind.

Single-reviewer gold sheet. The current gold sheet is rated by the lead author (WSET Diploma). We plan to expand to a panel of three WSET-Diploma-or-higher reviewers before the next release; inter-rater κ will then be reported as a first-class quality metric alongside agent-vs-human κ . The human-review web application (Appendix D) is designed for multi-reviewer IRR from the outset.

8 Ethics and Responsible Use

Alcohol context. Wine is an alcoholic beverage. OenoBench is a knowledge-evaluation tool, not a consumer-facing product. Models that score highly on OenoBench should not be deployed as drinking-recommendation systems without separate alignment work covering health information, age-gating, and jurisdiction-specific advertising rules.

Source licensing. Tier-1 government sources are public records. Tier-2 Wikipedia and Wikidata content is used under CC-BY-SA / CC0; OenoBench question texts are derivative works produced by paraphrasing facts, not copies of source paragraphs, and are released under **CC-BY-SA-4.0**. Per-fact source URLs are released alongside the corpus so downstream users can verify and re-attribute.

Personal data. The corpus contains entity names of producers, winemakers, and historical figures who are public-record principals of wine estates. No private individuals are identified. The gold-sheet reviewer is identified only by professional credential.

Misuse considerations. A wine-knowledge benchmark is low risk for direct misuse. The closest plausible misuse is using high-OenoBench-score models to automate fraudulent product descriptions or counterfeit provenance claims. We note this for completeness but do not see a benchmark-side mitigation distinct from general LLM-deployment safeguards. The construction code’s principal misuse risk is its application to retargeting toward high-stakes domains (medical, legal) without commensurate human-expert review of the retargeted source taxonomy; we discuss this scoping concern in Section 6.

LLM use during construction. Large language models were used as components of the pipeline (question generation, audit agents). They were *not* used as sources of factual claims; every factual assertion in the corpus traces to a non-LLM source URL with a tier-of-authority label. This separation is enforced by the A3 FactEcho agent (no verbatim source copying) and the closed-book pre-screen.

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A Datasheet for OenoBench

We follow the datasheet template of Gebru et al. [4].

A.1 Motivation

For what purpose was the dataset created? To evaluate factual knowledge of large language models on a multi-disciplinary, externally-validated domain (wine), and to provide a benchmark whose construction includes explicit bias controls (multi-model generation, multi-agent audit, self-preference scoring, closed-book vs. source-grounded partition).

Who created the dataset? The dataset was created by Nikita Khudov (StrategAI) for the NeurIPS 2026 Evaluations & Datasets Track. The lead author holds the WSET Diploma in Wines (the highest pre-Master-of-Wine qualification of the Wine & Spirit Education Trust) and produced the gold-sheet ratings used to calibrate the audit agents.

Funding. Independently funded by StrategAI. No external research grants were received. Open-Router API costs (\$236 cumulative across audit pilots and the final run, plus \$98 for the eval) were paid out-of-pocket.

A.2 Composition

What do the instances represent? Each instance is a multiple-choice benchmark question with: question text, 4 options with marked-correct answer, supporting fact(s) with source URL and tier label, generator model identifier, generation strategy, audit-agent verdicts (per-agent pass/warn/fail signals), and (for the gold-sheet subset) human-reviewer ratings on eight rubrics.

How many instances are there? 3,266 questions in `release_v1.2`. By strategy: fact-to-question 1,909, distractor mining 405, template 389, scenario 319, comparative 244. By difficulty (post-relabel): L1 694, L2 894, L3 678, L4 1,001. By domain: wine regions 1,108, grape varieties 766, producers 515, viticulture 502, wine business 250, winemaking 187. The corpus is built from 38,104 atomic facts across 35 scrapers and 580 unique source URLs.

What data does each instance consist of? Structured Postgres records mirroring the schema documented in Section 3 and Appendix B.

Recommended data splits. The dataset ships as a single test split (this is an evaluation-only benchmark). Two analytic partitions are documented and recommended for use: (i) the *closed-book* slice (1,601 questions tagged `closed_book_solvable`) and the *contextual* slice (1,665 remaining questions), partitioned by audit agent B2; and (ii) per-generator held-out splits for self-preference analysis (Section 5.3), defined dynamically by the `generator_family` column.

Errors, noise, redundancies? Documented in Section 4. Per-agent fail rates and κ versus the human gold sheet are released alongside the corpus. The 1,601 B2-flagged questions are kept with disclosure because LLM-judge calibration with humans on closed-book solvability is unreliable ($\kappa \approx 0.007$).

Self-contained, or relies on external resources? Self-contained: question text, supporting facts, and source URLs are all included. Source URLs may rot over time; we mitigate by archiving Tier-1 source pages and recording the access timestamp.

A.3 Collection process

How was the data acquired? By 35 provenance-verified web scrapers (released in `src/scrapers/`). Sources span government registries (INAO, TTB, OIV), inter-governmental bodies, university research groups (notably UC Davis, USDA Extension), Wikipedia, Wikidata, peer-reviewed journals (*OENO One*, *Vitis*, *AJEV*), and curated open datasets. Scraping was rate-limited and used the disclosed user agent `OenoBench-Research/1.0` (academic wine benchmark).

Time frame. Fact extraction: 2026-03 through 2026-04. Question generation: 2026-04 through 2026-05-03. Evaluation: 2026-05-03.

Ethical review. The work is non-IRB-eligible at our institution because (a) no human subjects were enrolled in research, (b) the gold-sheet reviewer is also the lead author, and (c) all source data is public-record information about commercial wine entities. Human review of LLM-generated questions is treated as data-quality assurance, not as human-subjects research.

A.4 Preprocessing / cleaning / labelling

Atomic-fact extraction with entity tagging; per-source paraphrase to avoid verbatim copy; near-duplicate suppression at cosine ≥ 0.92 ; nine-agent automated audit; human gold-sheet rating on a stratified 50-question subset per release.

A.5 Uses

Recommended uses. Evaluation of LLM factual knowledge in the wine domain; ablation studies of bias-mitigation techniques in benchmark construction; study of self-preference effects in LLM-as-judge pipelines; calibration of automated multi-agent QA frameworks against expert human review.

Discouraged uses. Direct deployment as a wine-recommendation or purchasing system without separate alignment, age-gating, and jurisdictional advertising review; fine-tuning on the benchmark itself for ranking purposes (would invalidate the benchmark for the trained model). The scrape-to-audit pipeline should not be retargeted to high-stakes domains (medicine, law) without comparable expert human review of the retargeted source taxonomy.

A.6 Distribution

Will the dataset be distributed? Yes, under **CC-BY-SA-4.0** on HuggingFace at <https://huggingface.co/datasets/oenobench/oenobench>. A Croissant manifest is included [1]. The construction code is on GitHub at <https://github.com/nikitahudov/oenobench>.

Citation.

```
@inproceedings{khudov2026oenobench,
  title = {OenoBench: A Wine-Domain Benchmark for Knowledge-Grounded
    Evaluation of Large Language Models},
  author = {Khudov, Nikita},
  booktitle = {Advances in Neural Information Processing Systems
    (NeurIPS), Datasets and Benchmarks Track},
  year = {2026}
```

}

A.7 Maintenance

Maintainer. Nikita Khudov, nikitahudov@gmail.com.

Update cadence. Wine regulation changes on a multi-year cadence; we plan annual re-extraction. Scraper code is released so the community can re-extract independently. Corpus versioning follows `release_vX.Y` (major release / minor patch) with all prior versions retained on HuggingFace.

Errata. A public errata log will be maintained alongside the HuggingFace release; corrections to questions, facts, audit verdicts, or difficulty labels will be versioned and dated. Reports of factual errors are accepted via GitHub issues and will be evaluated against the source URL.

B Additional Methodology Detail

B.1 Atomic fact extraction pipeline

The atomic-fact extractor (`src/scrapers/_fact_processing.py`) applies five sequential filters to each candidate sentence:

1. **Sentence split.** NLTK Punkt tokeniser; further split at explicit conjunctions (“and”, “but”, “while”) when both clauses have an independent verb. Maximum atomic length 30 words; reject longer.
2. **Reference resolution.** Pronouns and demonstratives are replaced with their entity referent from article context (paragraph-level coreference using a rule-based resolver tuned on Wikipedia leads). Sentences with unresolved *it/this/that* are dropped.
3. **Domain classification.** Lexicon-based + rule-based classifier maps each fact to one of the six domain pillars; ambiguous facts (no anchor term in any domain lexicon) are dropped.
4. **Length and predicate validation.** Reject facts <5 words, >50 words, missing a verb, or carrying dangling references / fragments. The 5–30 word band is the strict filter; the 31–50 band is kept with reduced confidence (910 facts).
5. **On-topic filter.** Region-specific keyword sets prevent cross-contamination (e.g. Austrian content in a Bordeaux scraper). Fail ⇒ drop.

A 2026-04 audit found that 19 of an original 35 scrapers contained hard-coded LLM-generated facts disguised as scraped data. The fix required rebuilding all 19 against live sources; the lesson preserved for future retargeting is that *provenance auditing must run on the scrapers themselves*, not only on the collected facts.

B.2 Wikidata SPARQL: P17 vs P131*

A common bug in early scrapers used the transitive administrative-parent property (P131*) to bind questions to a country. This introduced severe cross-region contamination, e.g. Austrian appellations appearing in a Bordeaux scraper because the SPARQL graph traversal walked through shared ancestor entities. Switching to the direct country relation (P17) eliminated cross-country leakage at the cost of ~3% lost coverage on borderline-region entities. The released SPARQL templates use P17 exclusively and we recommend the same discipline for any pipeline retargeted to a domain with hierarchical political subdivisions.

B.3 Question-generation prompts

The five strategies share a common JSON output schema enforced via Pydantic with a 3-tier extraction fallback (markdown-fenced, raw JSON, prefix-pattern). Per-strategy prompt summaries:

- **Fact-to-question:** "Given a single atomic wine fact, write a 4-option multiple-choice question whose unique correct answer is grounded in the fact. Paraphrase the fact; do not copy contiguous spans of >5 words. Distractors must be confusable same-category entities, not random alternatives."

- **Comparative:** given two related facts (matched on entity type, country/sub-domain), "ask which differs in the named attribute or what both share. Avoid iconic-vs-non-iconic pairings."
- **Scenario synthesis:** given a cluster of 3–5 facts, "write a brief domain-appropriate professional scenario (sommelier service / winemaker decision / viticulturist call / business/regulatory choice) with a unique correct answer that requires *all* the cluster facts to derive."
- **Distractor mining:** given a fact + 3 confusable entities, "write a question whose 3 wrong options are these entities, ranked by ascending plausibility."
- **Template:** 45 deterministic parameter substitution templates; no LLM call. A subset is paraphrased post-hoc by Gemini Pro to defeat the A4 stylometric fingerprint detector.

Full prompt texts are in `src/generators/_prompts.py`.

B.4 Closed-book gate specification

The closed-book pre-screen (`src/generators/_closed_book_gate.py`) runs a per-difficulty model:

- L1: Claude Haiku 4.5 reads the question with no source fact. Correct answer $\Rightarrow$ tag `closed_book_solvable`.
- L2: Claude Sonnet 4.6 same protocol.
- L3: Claude Opus 4.7 same protocol; the corpus quota cap fires here.
- L4: gate is skipped by protocol; expert items are expected to be answerable only with the source fact.

The gate’s role in generation is to enforce a corpus-level cap of 25% on closed-book-solvable items; the audit’s B2 agent later re-runs the check at corpus scale on a tri-judge panel. Failures are not dropped; they are reserved into the closed-book slice and used for the analytic partition in Section 5.5.

B.5 Audit agent specifications

Each agent is a versioned Python module under `src/qa/agents/`. Agents emit (`question_id`, `agent_id`, `version`, `severity`, `payload_json`) tuples to the `audit_findings` table; `severity` is one of {PASS, WARN, FAIL}. Versioning is strict: the (`run_id`, `question_id`, `agent_id`, `version`) tuple is unique, so re-running an agent at a new version produces additive findings rather than overwriting. We document each agent’s threshold and gold-sheet κ here.

Table 6: Audit-agent thresholds and gold-sheet calibration.

Agent	Signal	Threshold	κ vs. gold
A1 LexicalHygiene	vague/marketing regex hits	≥ 1 hit $\Rightarrow$ WARN; $\geq 3 \Rightarrow$ FAIL	0.71
A2 BiasStats	χ^2 position; M–W length	$p < 0.01 \Rightarrow$ FAIL	0.65
A3 FactEcho	LCS / fact length	$\geq 0.45 \Rightarrow$ WARN; $\geq 0.65 \Rightarrow$ FAIL	0.83
A4 TemplateFingerprint	POS-bigram logreg held-out AUC	> 0.84 corpus-level $\Rightarrow$ WARN	0.62
B1 TriJudgeAnswer	majority disagrees with key	2/3 disagree $\Rightarrow$ FAIL	0.78
B2 ClosedBookSolv.	majority solves no-source	2/3 solve $\Rightarrow$ WARN; 3/3 $\Rightarrow$ FAIL	0.007
C2 CategoryLeak	wine-type mismatch in distractors	any $\Rightarrow$ FAIL	0.91
C4 DifficultyAudit	$ \Delta $ assigned vs. rated	≥ 1 WARN; ≥ 2 FAIL	0.74
B3 UbiquityRisk	ubiquitous-grape stem $\times$ region answer	both $\Rightarrow$ WARN	0.69

The B2 row is the calibration finding discussed in Section 4.2: the LLM panel and the human reviewer disagree on what counts as “world-knowledge solvable”, because frontier LLMs have absorbed substantial wine knowledge during pre-training. We retain the agent as a measurement of the LLM panel’s prior, and report the contextual / closed-book partition (Section 5.5) that turns the disagreement into a useful diagnostic.

C Full Evaluation Tables

We give the full per-domain, per-strategy, and per-difficulty breakdowns referenced in Section 5, plus the cost ledger and the post-evaluation zero-correct audit.

Table 7: Per-config per-domain accuracy (%).

Config	Regions	Varieties	Producers	Viticulture	Winemaking	Business
claude-opus-4.7	83.2	80.2	86.8	77.9	86.6	64.2
claude-haiku-4.5	55.8	49.7	58.7	54.2	56.7	37.8
gpt-5	88.1	78.2	88.4	82.8	80.7	63.0
gpt-5-mini	82.1	74.4	83.5	77.5	77.5	66.3
gemini-2.5-pro	87.0	78.8	89.0	78.5	77.5	61.8
gemini-2.5-flash	76.9	72.9	82.3	73.6	77.5	60.6
llama-3.3-70b	68.9	62.9	77.2	65.7	70.6	50.4
llama-3.1-8b	62.2	55.6	69.5	61.7	59.4	47.6
deepseek-v3	74.4	68.5	73.2	70.8	69.0	51.2
qwen-2.5-72b	69.0	65.1	73.4	67.5	70.1	52.8
qwen-2.5-7b	59.1	55.6	59.4	57.4	57.8	44.7
mistral-large-2411	71.5	65.6	75.4	69.4	73.3	51.6
o3	88.4	81.3	90.9	80.3	76.5	65.4
gemini-2.5-pro-thinking	86.5	80.5	90.0	80.5	83.4	60.2
deepseek-r1	80.5	73.6	85.6	72.2	76.5	64.6
claude-opus-4.7-thinking	83.2	80.9	87.0	78.7	84.0	61.4
Mean (16 configs)	76.0	70.2	79.4	71.8	73.6	56.5

Table 8: Per-config per-strategy accuracy (%).

Config	FTQ	Scenario	Template	Comparative	Distractor
claude-opus-4.7	77.7	85.6	92.8	79.5	82.7
claude-haiku-4.5	47.5	62.7	70.7	53.3	56.8
gpt-5	79.9	80.9	95.4	79.5	87.9
gpt-5-mini	73.8	80.6	92.8	78.3	84.7
gemini-2.5-pro	77.4	87.5	93.1	82.4	86.4
gemini-2.5-flash	69.5	84.0	89.7	77.5	79.5
llama-3.3-70b	59.5	80.6	89.7	64.8	71.9
llama-3.1-8b	52.2	70.5	89.7	60.2	63.7
deepseek-v3	64.5	80.3	86.4	71.7	73.3
qwen-2.5-72b	59.6	79.3	88.7	69.3	73.6
qwen-2.5-7b	49.1	72.1	78.9	57.0	61.0
mistral-large-2411	62.7	80.6	87.4	67.6	73.3
o3	79.9	84.3	94.9	84.8	88.4
gemini-2.5-pro-thinking	78.5	88.4	95.6	84.0	84.2
deepseek-r1	73.4	75.5	92.5	77.0	80.7
claude-opus-4.7-thinking	77.3	85.6	93.6	80.3	83.0
Mean (16 configs)	67.6	79.9	89.5	73.0	76.9

C.1 Item analysis

C.2 Self-preference bootstrap CIs

D Application Screenshots

This appendix shows screenshots of two interfaces released alongside the corpus: the human-review web application (`src/review_app/`, port 5556) used to collect WSET-Diploma gold-sheet ratings, and the project monitoring dashboard (`src/dashboard/`, port 5555) used during construction.

D.1 Human-review web application

The review app is a Flask application with two layers of authentication (outer HTTP Basic for shared-link access, inner per-reviewer session cookies) and writes ratings into a separate `human_reviews` Postgres table to keep gold-sheet labels independent of automated audit findings. Each reviewer is associated with a WSET certification level recorded at registration; the app supports multi-reviewer IRR (κ) computation out of the box.

Table 9: Per-config per-difficulty accuracy (%).

Config	L1 (entry)	L2	L3	L4 (expert)
claude-opus-4.7	98.0	77.0	86.7	69.1
claude-haiku-4.5	74.3	48.3	59.9	38.8
gpt-5	98.3	82.4	87.5	69.2
gpt-5-mini	98.0	73.7	84.7	64.8
gemini-2.5-pro	98.6	79.2	87.3	68.5
gemini-2.5-flash	95.4	70.9	82.2	60.1
llama-3.3-70b	91.9	60.6	74.8	50.3
llama-3.1-8b	84.8	55.4	65.3	45.0
deepseek-v3	95.8	65.9	76.3	52.4
qwen-2.5-72b	93.2	60.5	73.7	51.4
qwen-2.5-7b	82.0	50.6	60.0	43.3
mistral-large-2411	94.4	63.1	73.3	54.0
o3	98.4	81.4	89.4	71.2
gemini-2.5-pro-thinking	99.1	80.2	88.3	69.4
deepseek-r1	97.5	74.2	82.0	62.1
claude-opus-4.7-thinking	98.3	76.8	87.0	68.6
Mean (16 configs)	93.6	68.8	78.7	58.7

Table 10: Reasoning lift per difficulty tier (δ = thinking – standard, pp).

Pair	L1	L2	L3	L4
Claude Opus 4.7 (think vs std)	+0.3	−0.2	+0.3	−0.5
Gemini 2.5 Pro (think vs std)	+0.5	+1.0	+1.0	+0.9
OpenAI o3 vs GPT-5	+0.1	−1.0	+1.9	+2.0
DeepSeek R1 vs V3	+1.7	+8.3	+5.7	+9.7

D.2 Monitoring dashboard

D.3 HuggingFace dataset card

E Released Artifacts and Reproducibility

E.1 Released artifacts

- **Dataset (HuggingFace).** https://huggingface.co/datasets/oenobench/oenobench—release_v1.2, 3,266 questions, Parquet test split, README.md datasheet, croissant.json manifest. License: CC-BY-SA-4.0. Versioning is permanent: prior releases remain accessible via revision tags.
- **Code (GitHub).** <https://github.com/nikitahudov/oenobench> — full pipeline: 35 scrapers, 5 question generators, 9 audit agents, the human review web app, and the monitoring dashboard. Apache 2.0.
- **Pipeline diagrams.** Included in docs/figures/ as Mermaid (.mmd), PNG, and SVG.
- **Process log.** docs/PROCESS_LOG.md — chronological lab-notebook of every phase shipped, with sources, methodology, quality-control counts, and decision rationale, in the format prescribed for the methodology sections of this paper.
- **Audit reports.** docs/QUALITY_AUDIT_REPORT.md, docs/RELEASE_V1_2_AUDIT_ACTIONS.md, docs/GOLD_CALIBRATION_ANALYSIS.md, data/reports/zero_correct_97_audit.md.
- **Eval reports.** data/reports/eval_release_v1_2_full_cleaned.md (the 16-config × 3,266-Q run reported in this paper).

E.2 Reproducibility scripts

The construction pipeline is driven by a small number of shell scripts in scripts/:

- `run_all_scrapers.sh` — orchestrates the 35 scrapers end-to-end. Each scraper writes a timestamped log to data/logs/.

Table 11: Cost ledger and cost-per-correct-answer.

Config	Total cost	Correct ans.	Cost / correct
llama-3.1-8b	\$0.01	1,976	\$0.0000
llama-3.3-70b	\$0.04	2,190	\$0.0000
gemini-2.5-flash	\$0.12	2,454	\$0.0000
deepseek-v3	\$0.12	2,295	\$0.0001
qwen-2.5-7b	\$0.12	1,860	\$0.0001
qwen-2.5-72b	\$0.15	2,202	\$0.0001
claude-haiku-4.5	\$0.44	1,741	\$0.0003
mistral-large-2411	\$0.87	2,256	\$0.0004
gpt-5-mini	\$2.82	2,561	\$0.0011
claude-opus-4.7	\$3.35	2,647	\$0.0013
claude-opus-4.7-thinking	\$3.53	2,645	\$0.0013
deepseek-r1	\$10.54	2,517	\$0.0042
o3	\$11.80	2,729	\$0.0043
gemini-2.5-pro-thinking	\$13.06	2,698	\$0.0048
gpt-5	\$21.90	2,704	\$0.0081
gemini-2.5-pro	\$29.47	2,669	\$0.0110

Table 12: Per-config Self-Preference Score with 95% bootstrap CI (1,000 resamples). DeepSeek and Mistral have no own-family questions in release_v1.2 and are omitted.

Config	Family	own (%)	other (%)	δ (pp)	95% CI
claude-haiku-4.5	anthropic	58.8	48.8	+10.0	[+5.7, +14.5]
claude-opus-4.7	anthropic	86.6	77.5	+9.1	[+6.1, +12.1]
claude-opus-4.7-thinking	anthropic	86.9	77.2	+9.7	[+6.5, +13.1]
qwen-2.5-7b	qwen	60.9	51.9	+9.0	[+4.5, +13.2]
qwen-2.5-72b	qwen	66.1	64.1	+2.0	[−2.1, +6.0]
llama-3.3-70b	meta	65.5	63.6	+1.9	[−2.3, +6.1]
gpt-5-mini	openai	77.9	76.1	+1.7	[−2.4, +5.4]
o3	openai	81.9	82.1	−0.1	[−3.9, +3.3]
gpt-5	openai	79.7	81.4	−1.7	[−5.4, +2.1]
llama-3.1-8b	meta	54.2	57.2	−3.0	[−7.3, +1.4]
gemini-2.5-pro	google	75.0	81.1	−6.1	[−10.7, −1.8]
gemini-2.5-pro-thinking	google	74.0	82.0	−8.0	[−12.3, −3.8]
gemini-2.5-flash	google	64.3	74.7	−10.4	[−15.3, −5.6]

- `run_release_v1_build.sh` — full generation pipeline invocation; idempotent and resume-safe via stamped tags.
- `export_release_v1_2_to_parquet.py` — produces the HuggingFace Parquet directly from Postgres; the released file is the byte-identical output of this script.
- `build_smart_review_sheet.py` — generates a 50-question stratified gold-sheet review batch (random + critical-FAIL + borderline-WARN strata).
- `tag_audit_actions.py` — applies the audit drop / relabel policy to a tagged corpus; this is the script that produced the release_v1.2 state from release_v1.1.

E.3 Database schema

The released schema is initialised by the migrations under `config/postgres/` (six idempotent files, applied in order on container startup): `init.sql` (sources, facts, questions, question_facts, generation_metadata, evaluation_runs, evaluation_answers, human_reviews), `002_audit_schema.sql` (audit_runs, audit_findings, audit_gold_labels, audit_severity enum), `003_sample_schema.sql` (mirror schema for the Phase 5 sample-DB eval), `003_cb_reserve.sql` (closed-book reserve pool tagging), `004_eval_telemetry.sql` (provider, tokens, reasoning_config, latency columns on evaluation_answers), `005_or_cost_telemetry.sql` (OpenRouter authoritative cost columns).

A simplified entity-relationship diagram is in Figure 19.

E.4 Compute and cost

The full evaluation reported in Section 5 ran in **120 minutes 34 seconds** on a single VM driving 16 OpenRouter configurations in parallel. Total LLM API spend across construction was approximately:

- Generation pipeline: \$160 (across audit pilots v1–v16 and the final `release_v1` build).
- Audit pipeline (`release_v1.2` cycle): \$76.
- Evaluation (16 configs $\times$ 3,266 Q): \$98.33 effective OpenRouter cost.

No GPU compute was used; all model inference was via OpenRouter (a unified API gateway). The release artefacts include all per-call telemetry (provider, tokens, reasoning configuration, p50/p95 latency, OR-authoritative cost) for the full evaluation run.

E.5 Test status

771 of 771 unit tests pass on `main` as of the camera-ready build. Tests cover the fact-processing pipeline, the five generation strategies, all nine audit agents, the orchestrator, the closed-book gate, the eval harness (slot dispatch, cost computation, reasoning-mode prompt assembly), and the review-app endpoints.

E.6 Anonymisation note

For double-blind submission, the public-facing URLs in this paper are mirrored to anonymous fora as documented in `docs/ANONYMIZATION_PLAN.md`. The HuggingFace dataset is mirrored at an anonymous handle, the GitHub repository is exposed via `anonymous.4open.science`, and all author/funding declarations are blanked in the OpenReview submission. The camera-ready will swap all anonymous URLs back to their canonical equivalents; the mapping is preserved in the anonymisation plan file in the released code.

F Construction Timeline

The full chronological lab notebook of construction is in `docs/PROCESS_LOG.md`. We summarise the major dated milestones here for paper-traceable provenance.

Table 13: Construction-and-evaluation timeline (selected milestones).

Phase	Dates	Milestone
1	2026-03-12 – 04-01	35 scrapers built; first 16,702-fact corpus assembled.
1.1	2026-04-07	<i>Provenance audit reveals 19 scrapers contained hardcoded LLM-generated facts</i> ; corpus paused.
1.2	2026-04-08 – 04-11	Shared infrastructure (<code>_fact_processing.py</code> , <code>_web_helpers.py</code> , <code>_wiki_helpers.py</code>) shipped; Phase 1 rebuild fixes 8 quality-issue scrapers; Phase 2 rebuild replaces all 17 hardcoded scrapers with genuine Wikipedia + SPARQL + official sources. <i>38,104 fact corpus restored, all provenance-traceable</i> .
2	2026-04-12 – 04-22	Five generation strategies built and iteratively tuned; multi-model orchestrator with quota management; closed-book gate v2 wired into all five generators.
2c	2026-04-19 – 05-02	Nine-agent audit framework shipped across 4 teams; sixteen audit pilots (v1–v16) tune thresholds and speedups; 472/472 $\rightarrow$ 771/771 tests pass.
2g	2026-04-28 – 05-02	Ten generation-pipeline speedup levers shipped (LLM cache, in-process dispatch, Thread-Pool dispatch, substantiveness filter, circuit breaker, tier-aware gate). v9 build wall: 18 min vs. v8’s 11.4 h.
2j	2026-05-03	<code>release_v1.1</code> (3,670 candidate questions) assembled from staged generation runs and the curated sample-DB. Audit cycle drops 341 questions on critical FAILs and re-labels difficulty for 1,259 (51% L3+L4 vs. prior 14%). Cycle cost \$76 / 5h 22m.
4	2026-05-03	Human-review web app shipped on port 5556 with separate <code>human_reviews</code> schema and multi-reviewer IRR support.
5	2026-05-03 – 05-04	Full evaluation: 16 configs $\times$ 3,329 questions in 120m 34s for \$98.33; post-eval audit of 97 zero-correct items removes 54 defects + 9 borderline-review drops; <code>release_v1.2</code> locked at 3,266 questions.
HF	2026-05-04	Dataset card + Croissant manifest + Parquet test split published; this paper drafted.

The timeline contains one decisive contingency that is methodologically worth flagging: the 2026-04-07 provenance audit, which discovered that about half the original scrapers had been silently substituting LLM-generated content for live source data, was the single most expensive course-correction in the project (more than four days of parallelised rebuild). It is the reason every fact-collection step in the final pipeline (Section 3) emits a per-source URL trace check, and it is the reason we recommend in Section 6 that any retargeting of the pipeline to another expert domain run a comparable scraper-side provenance audit before publication.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: Section 1 lists four contributions, and each is supported by a numbered section: dataset construction (Section 3), automated audit (Section 4), bias-aware evaluation (Section 5). The abstract’s headline numbers (3,266 questions, 16-config eval, +32.6 pp closed-book vs. contextual gap) are reproduced with full tables in Appendix C.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 7 discusses six limitations explicitly, including English-only scope, geographic skew, snapshot vs. moving target, B2 audit calibration ceiling, residual self-preference, and single-reviewer gold sheet.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This is a dataset-and-benchmark paper; no formal theoretical results are claimed.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper?

Answer: [Yes]

Justification: Appendix E lists the construction scripts, the dataset URL, the model slate, the per-call telemetry available alongside the released corpus, and the deterministic build pipeline. Section 5 reports run id, wall time, total cost, and exact config strings.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Code at <https://github.com/nikitahudov/oenobench> (Apache 2.0); data at <https://huggingface.co/datasets/oenobench/oenobench> (CC-BY-SA-4.0). Exact reproduction commands are documented in Appendix E.

6. Experimental setting/details

Question: Does the paper specify all the training and test details necessary to understand the results?

Answer: [Yes]

Justification: Section 5 specifies the 16-config slate, single-letter output protocol (A–D, max_tokens=5, five-stop fallback). Appendix C gives full per-config × per-domain × per-strategy × per-difficulty tables. Appendix B gives audit-agent thresholds and prompt summaries.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Reasoning lift, self-preference scores, and closed-book vs. contextual deltas are all reported with 95% bootstrap confidence intervals (1,000 resamples), with the bootstrap procedure documented in Section 5.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

Answer: [\[Yes\]](#)

Justification: Appendix E reports total wall time (120 min 34 s), total LLM calls (52,256), total cost (\$98.33 effective), and the deployment topology (single VM driving 16 OpenRouter configurations in parallel; no GPU compute).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [\[Yes\]](#)

Justification: All sources used are public-record; no human subjects research was conducted (the gold-sheet reviewer is the lead author); responsible-use guidance is given in Section 8.

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [\[Yes\]](#)

Justification: Section 8 discusses the alcohol-context, source-licensing, personal-data, and misuse considerations; Section 6 discusses positive applications including specialist-model fine-tuning and pipeline retargeting to other expert domains.

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

Answer: [\[Yes\]](#)

Justification: The dataset is text-only multiple-choice questions with public-record entity names; no scraped images or personal-data risks. We document discouraged uses (Section 8, Appendix A) including the recommendation against retargeting the pipeline to high-stakes domains without expert review.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [\[Yes\]](#)

Justification: Source-tier table in Section 3.2 credits all major sources with licenses (Wikipedia CC-BY-SA, Wikidata CC0, INAO Licence Ouverte, TTB public domain, etc.). Per-fact source URLs are released alongside the corpus.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [\[Yes\]](#)

Justification: A full Datasheet is in Appendix A; a Croissant manifest [1] ships with the HuggingFace release; all 9 audit agents, 5 generators, and 35 scrapers are documented in source.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation?

Answer: [\[N/A\]](#)

Justification: No crowdsourcing or third-party human-subjects research. Gold-sheet ratings were produced by the lead author (a WSET-Diploma holder); the review web app screenshots in Appendix D document the rubric and instructions.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether IRB approvals were obtained?

Answer: [N/A]

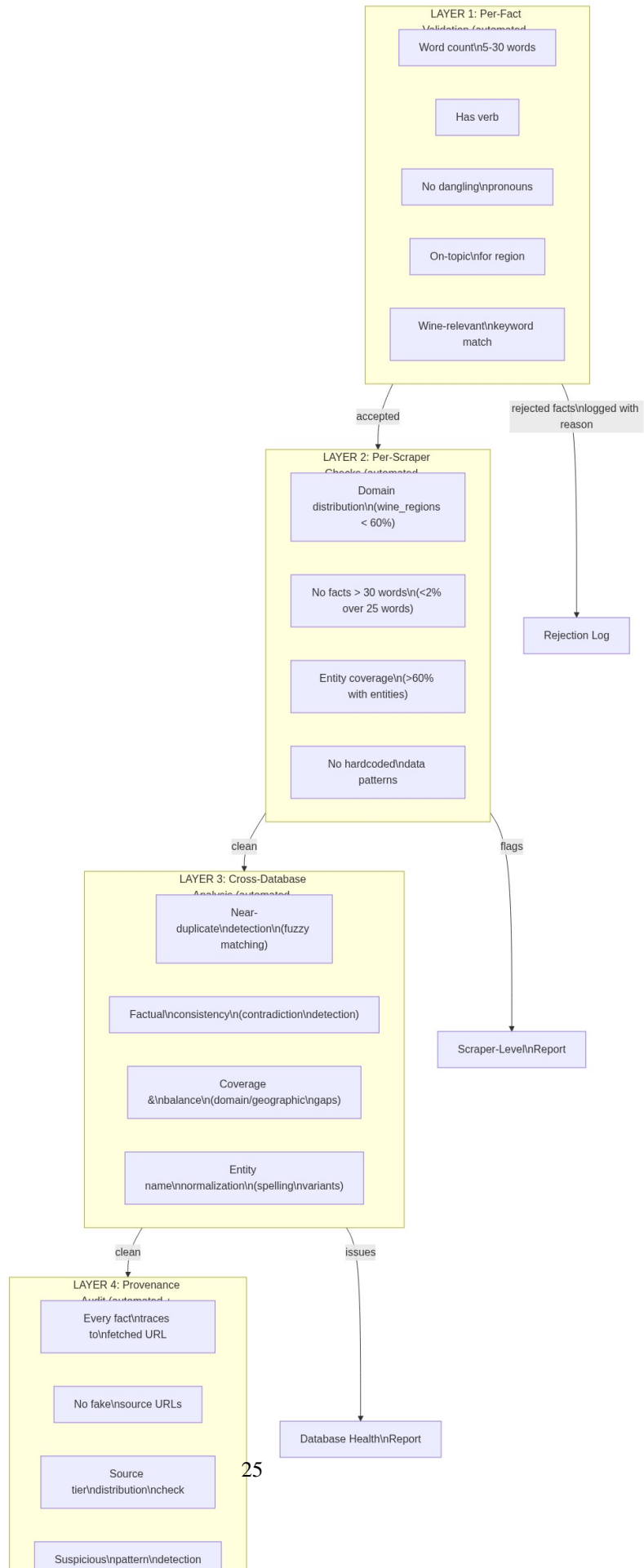
Justification: As noted in Appendix A, the work involves no human subjects research distinct from data-quality assurance by the lead author.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: [Yes]

Justification: LLMs are central to the methodology (multi-strategy question generation across five model families, nine-agent automated audit including a tri-judge panel, closed-book solvability pre-screen). Section 3 and Section 4 describe their usage; Section 8 and Appendix B document the explicit fact-grounding constraint that prevents LLMs from being used as the source of factual claims.



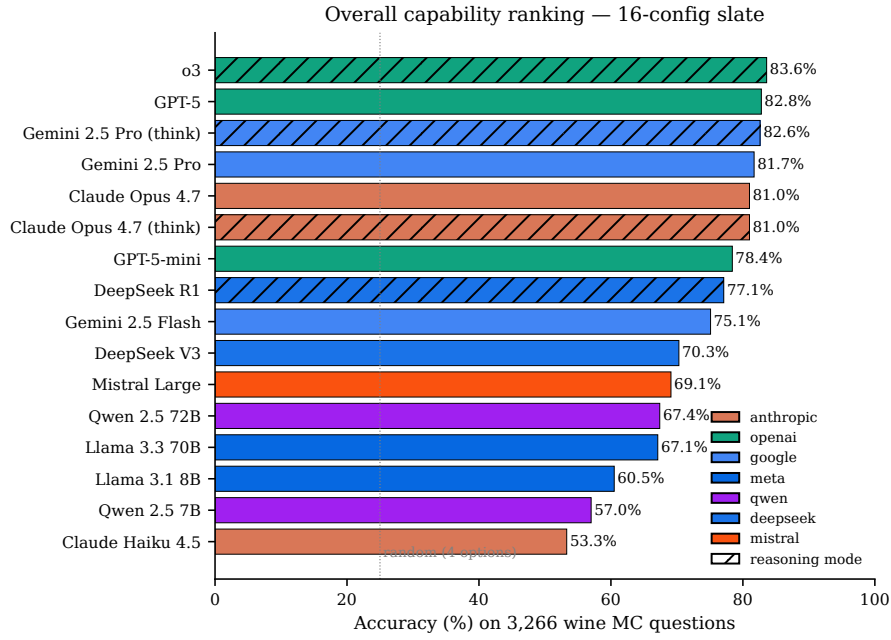


Figure 4: Overall accuracy on the 3,266-question corpus across 16 configurations. Reasoning-mode configs are hatched. Random-baseline (4-option MC) at 25% is shown as a dotted line.

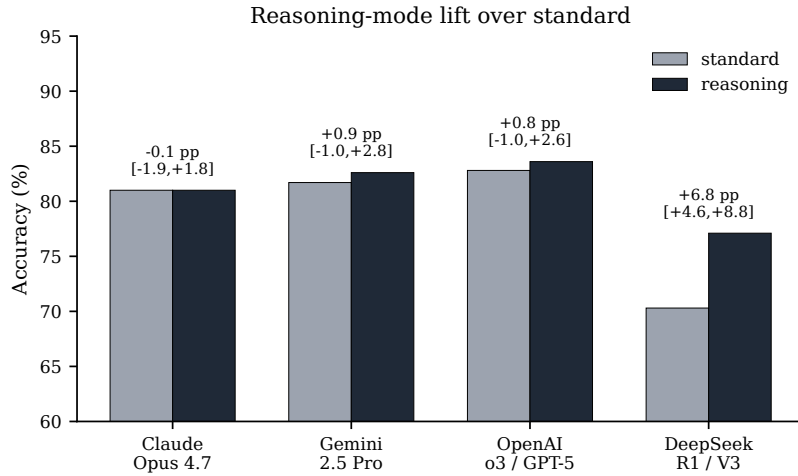


Figure 5: Accuracy of reasoning-mode vs. standard configurations for four families. Numbers above bars: δ in pp with bootstrap 95% CI from 1,000 resamples. Only DeepSeek shows a statistically significant lift; for the other three, zero is inside the CI.

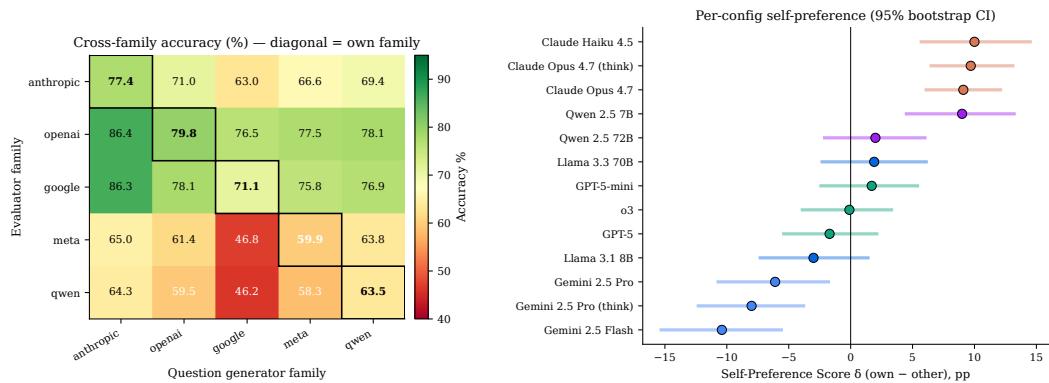


Figure 6: Self-preference bias. *Left*: cross-family accuracy matrix; rows are evaluator family, columns are question-generator family, diagonal entries are own-family. *Right*: per-config Self-Preference Score $\delta = \text{Acc}(\text{own}) - \text{Acc}(\text{other})$ with 95% bootstrap CI.

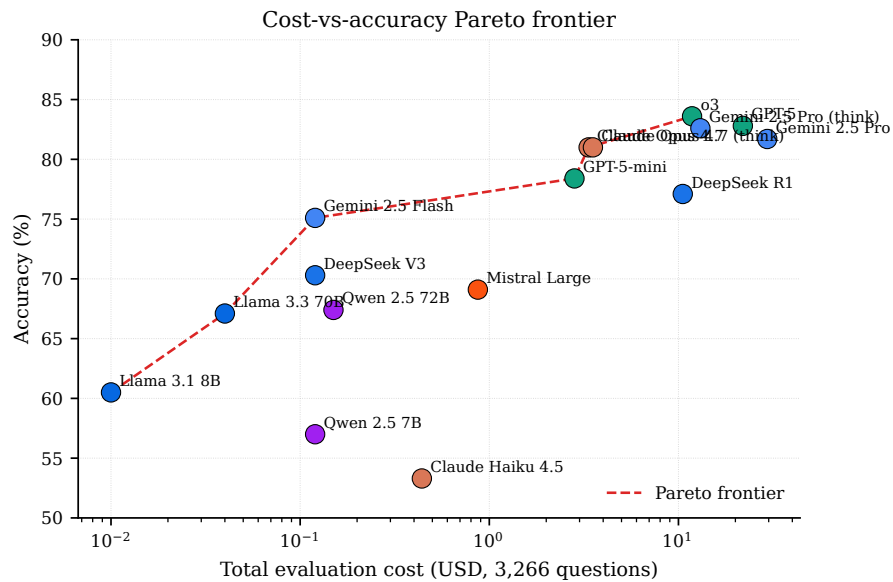


Figure 7: Total OpenRouter cost (log scale) versus accuracy on the full 3,266-question slate. Pareto frontier (red dashed) illustrates the cost-vs-quality trade-off: Llama-8B and Llama-70B anchor the cheap end, GPT-5-mini at \$2.82 dominates everything cheaper, and o3 at \$11.80 dominates everything more accurate.

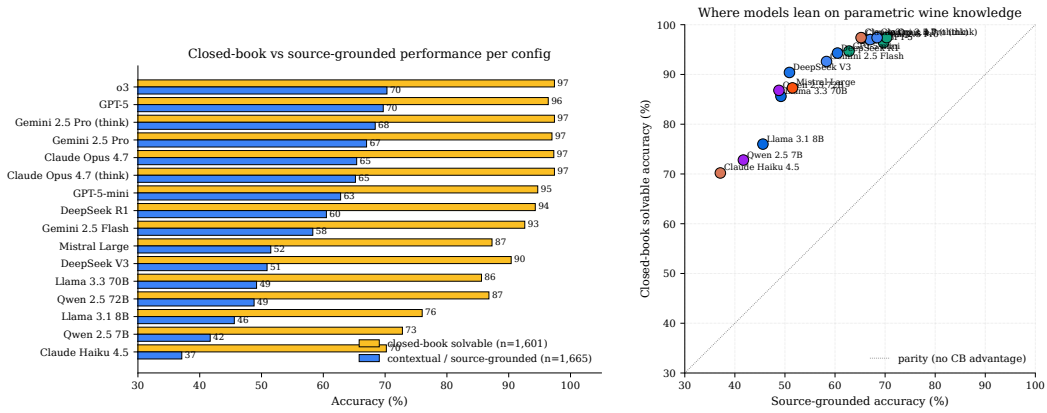


Figure 8: Performance partitioned by audit verdict. *Left*: Per-config accuracy on B2-flagged closed-book-solvable questions ($n=1601$, yellow) vs. contextual / source-grounded questions ($n=1665$, blue). *Right*: same data as a scatter; the diagonal is parity. Every config sits well above parity, with δ ranging from +26.6 pp (GPT-5) to +39.6 pp (DeepSeek V3).

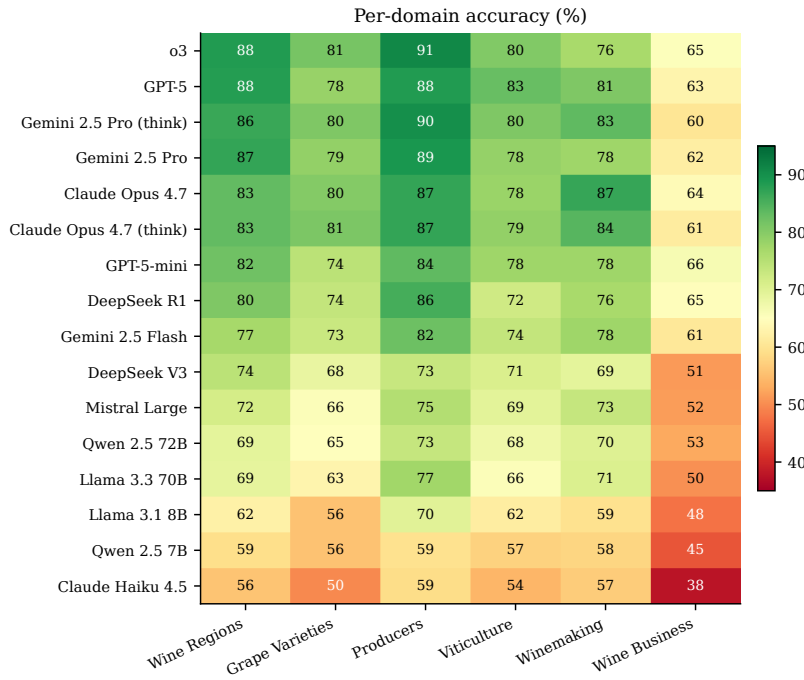


Figure 9: Per-domain accuracy as a heatmap. *Wine business* is the hardest domain across every tier (mean 56.5%) — both because business-domain facts are sparser in the training distribution and because the C4 difficulty re-label promoted business questions that are computationally harder (multi-fact reasoning over regulation).

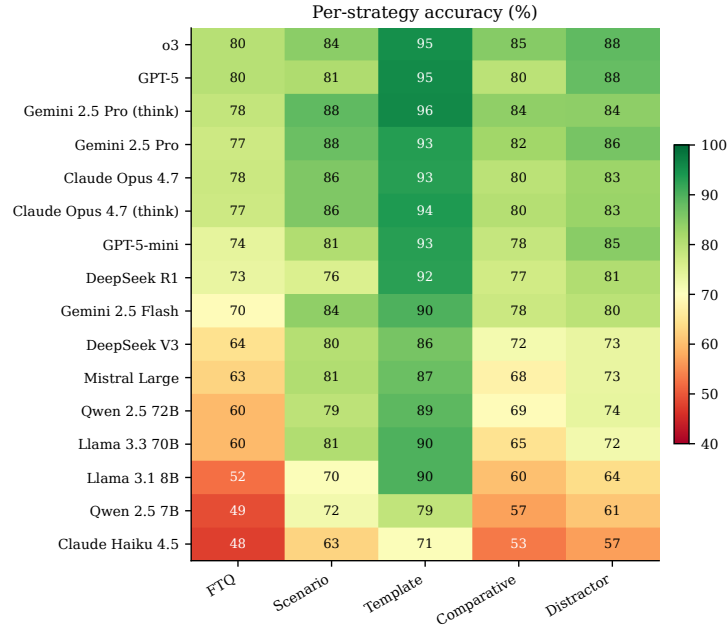


Figure 10: Per-strategy accuracy heatmap. Templates are the easiest strategy uniformly (89.5% mean), even for small models, because they are deterministic single-fact recall questions. The fact-to-question strategy is hardest because its distractors are sampled from confusable same-category entities; comparative questions add the constraint that the model must integrate two facts.

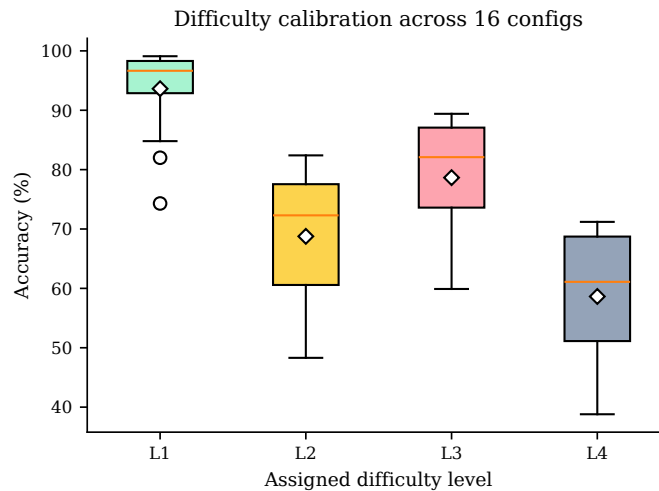


Figure 11: Difficulty calibration: distribution of accuracy across the 16 configs at each level. The medians sit at L1 96%, L2 70%, L3 81%, L4 60%. The L2<L3 inversion is an artifact of the C4 re-label: many questions originally tagged L2 by their generator were promoted to L3, leaving the residual L2 set comprising items that were correctly tagged in the first place — a smaller-but-truer L2 partition.

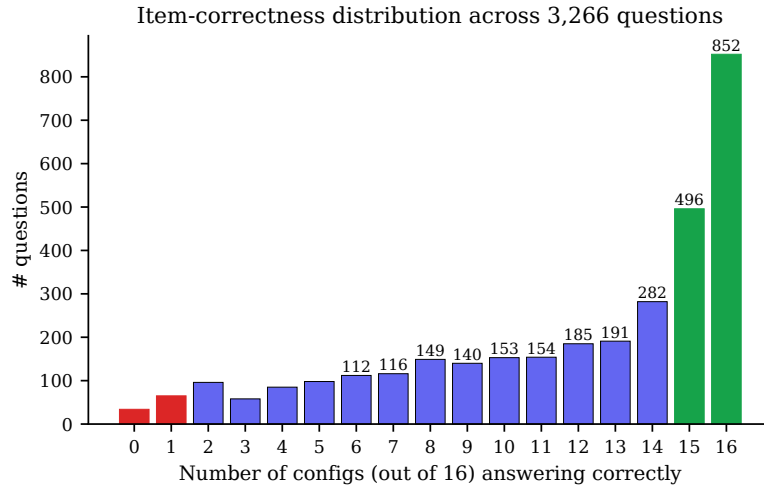


Figure 12: Item-correctness distribution. The right tail ($k \geq 15$ correct) holds 1,348 ceiling items (41%); the left tail ($k \leq 1$) holds 99 items, 97 of which were audited post-eval. The bulk of the mass sits in the discriminating middle ($k=4-14$).

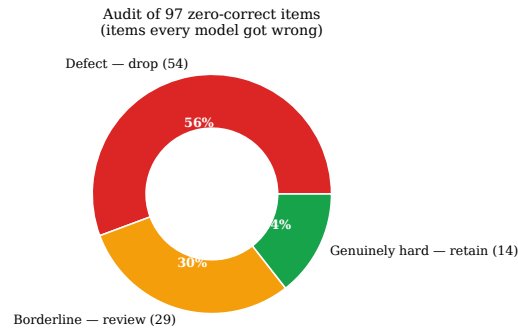


Figure 13: Audit of 97 zero-correct items. We reviewed all 97 questions every model answered incorrectly: 54 were genuine corpus defects (wrong key, all-correct options, duplicate or equivalent options), 29 were borderline cases that warranted further review (and were dropped after a second-pass review on 2026-05-04), and 14 were retained as legitimately hard. The defect rate (56% among zero-correct items) underlines the value of an empirical post-eval audit alongside automated agents.

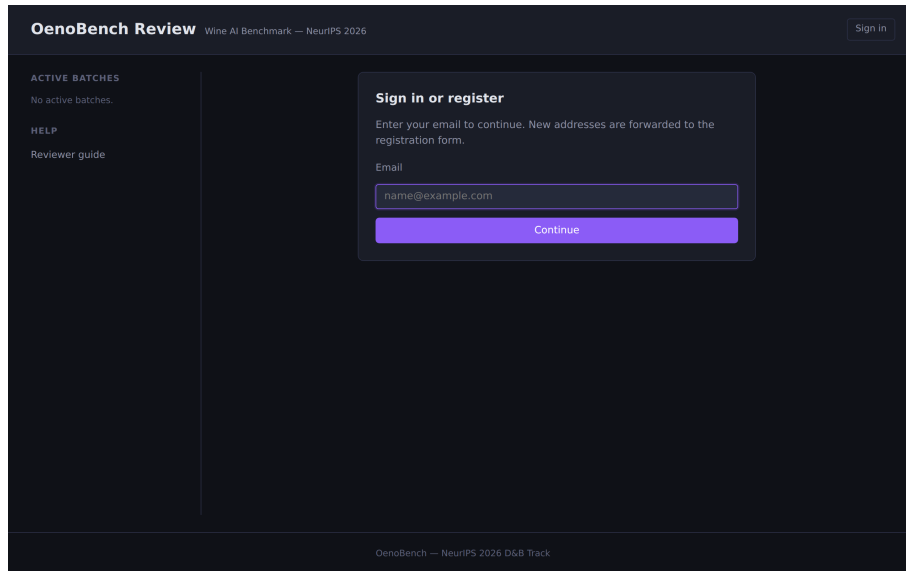


Figure 14: Sign-in / register screen. Outer HTTP Basic Auth gates shared-link access; the inner form establishes a per-reviewer session for IRR attribution.

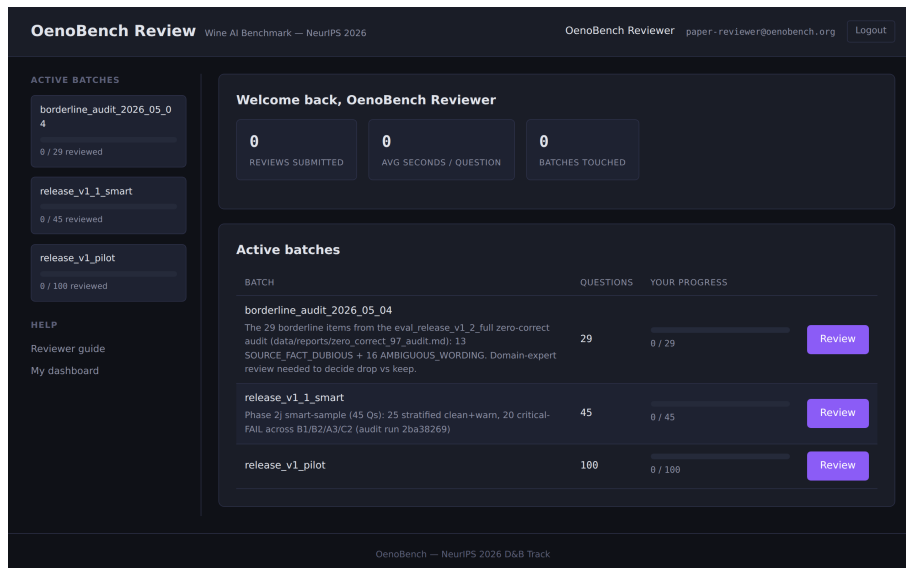


Figure 15: Reviewer dashboard. Active batches are listed; clicking a batch enters the rubric-scoring flow. The reviewer guide is linked permanently in the sidebar.

ACTIVE BATCHES

borderline_audit_2026_05_04

0 / 29 reviewed

release_v1_1_smart

0 / 45 reviewed

release_v1_pilot

0 / 100 reviewed

HELP

Reviewer guide

My dashboard

borderline_audit_2026_05_04 / WB-GRP-0388-L3

0 / 29 reviewed

Back to dashboard

Difficulty 3

analysis

multiple-choice

grape-varieties

Which grape variety is authorized for cultivation in the Ramona Valley AVA, known for its warm climate and focus on Mediterranean varieties?

- A Grenache
- B Mourvèdre
- C Carignane ✓ KEYED
- D Syrah

Source facts

[1] Ramona Valley AVA permits the Carignane grape variety.

[2] North Yuba AVA permits the Grenache grape variety.

Rubric scoring

Show definitions

Click Pass, Warn, Fail, or leave a row unset (Skip).
1 / 2 / 3 / 4 chips on the focused row, Tab / Shift + Tab move row, V opens verdict, Enter submits, Esc clears row.

Does the keyed answer correspond to the wine fact in the source?

Answer correct ?

Pass Warn Fail Skip

Pass: Keyed answer is unambiguously correct given the source.
Warn: Technically defensible but the source is ambiguous.
Fail: Keyed answer contradicts the source, or the source does not support it.

Suggested: A-D

Are the wrong options plausible, close, with no category leak?

Distractors plausible ?

Pass Warn Fail Skip

Pass: Each distractor sits in the same wine category as the keyed answer and could fool a reasonable taker.
Warn: One distractor is weak or its category leaks the answer, but the question is still non-trivial.
Fail: Two or more distractors are obviously wrong, or the categorical split eliminates them on sight.

Is there exactly one defensible answer among the four options?

Not ambiguous ?

Pass Warn Fail Skip

Pass: Exactly one defensible answer.
Warn: A second option is technically defensible only on a wine-trivia stretch.
Fail: Two or more options are equally well supported by mainstream wine references.

Does the question stay within the semantic claims of the source fact(s)?

Source faithful ?

Pass Warn Fail Skip

Pass: Question, options, and explanation only assert claims supported by source facts.
Warn: One minor factual extension you happen to know is true but isn't in source_facts.
Fail: The question makes a wine claim not supported by the listed source facts.

Could a knowledgeable wine taker answer without consulting the source?

Needs source ?

Pass Warn Fail Skip

Pass: Question genuinely requires the source — the specific datum is needed.
Warn: A wine pro can guess but a layperson cannot.
Fail: Answer is part of basic wine knowledge (e.g. Nebbiolo for Barolo).

Is the stem free of marketing fluff and hedge-words?

No vague language ?

Pass Warn Fail Skip

Pass: Clean, neutral, factual stem.
Warn: One borderline phrase but the question is still answerable.
Fail: Stem is so hedged or florid the answer becomes a matter of interpretation.

Do the difficulty (1-4) and cognitive labels match the

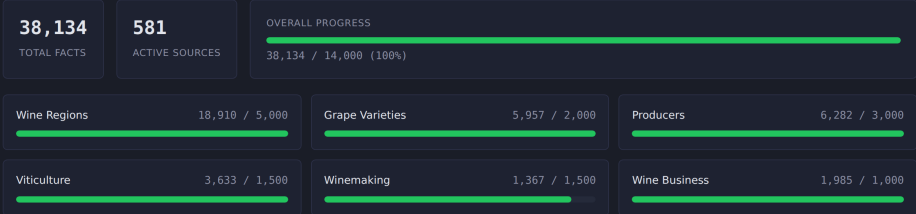
Labels correct ?

Pass Warn Fail Skip

Project Overview



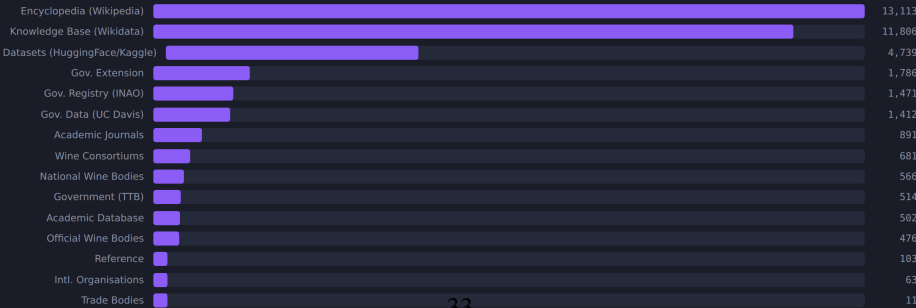
Fact Collection PHASE 1 COMPLETE



Country × Domain Distribution

COUNTRY	WINE REGIONS	GRAPE VARIETIES	PRODUCERS	VITICULTURE	WINEMAKING	WINE BUSINESS	TOTAL
General	2,903	1,430	2,795	2,365	948	1,423	11,864
Portugal	5,755	62	110	217	5	27	6,176
United States	2,268	870	596	345	37	61	4,177
France	2,666	226	365	84	73	23	3,437
Italy	1,755	890	178	188	88	69	3,088
Germany	555	485	395	107	36	81	1,659
Spain	388	327	714	57	25	32	1,543
Australia	417	391	257	104	20	58	1,247
South Africa	241	347	139	47	44	61	879
New Zealand	217	280	103	75	18	30	723
Slovenia	522	28	10	7	4	6	577
Austria	246	37	180	6	7	9	485
Canada	108	115	186	24	15	32	480
Hungary	212	158	19	41	17	14	461
England	163	70	50	29	10	39	361
Georgia	150	145	24	10	16	10	355
Greece	227	37	7	—	—	1	272
Croatia	76	47	5	9	4	6	147
Chile	39	5	59	—	—	1	104
Argentina	27	5	10	—	—	1	43
Israel	8	4	11	—	—	1	24
Lebanon	—	—	2	—	—	—	2
Total	18,943	5,959	6,215	3,635	1,367	1,985	38,104

Source Type Distribution



Question Generation RELEASE_V1.2



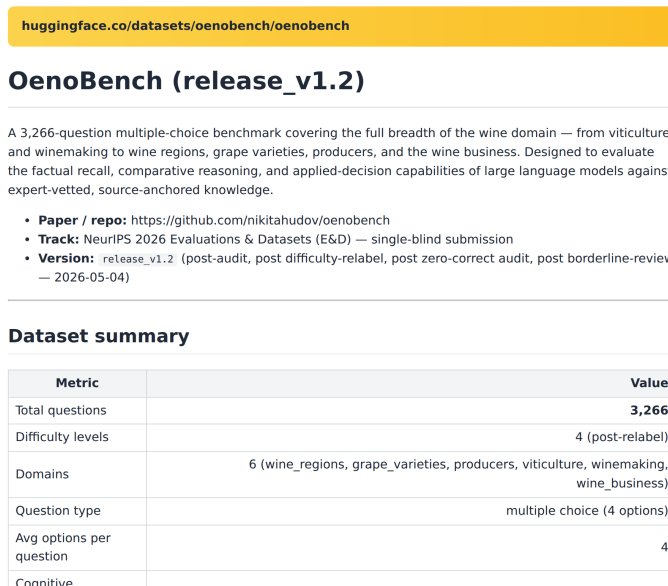


Figure 18: HuggingFace dataset card (preview). The release ships a full datasheet, a Croissant manifest [1], the 3,266-question Parquet test split, and per-fact source URLs.

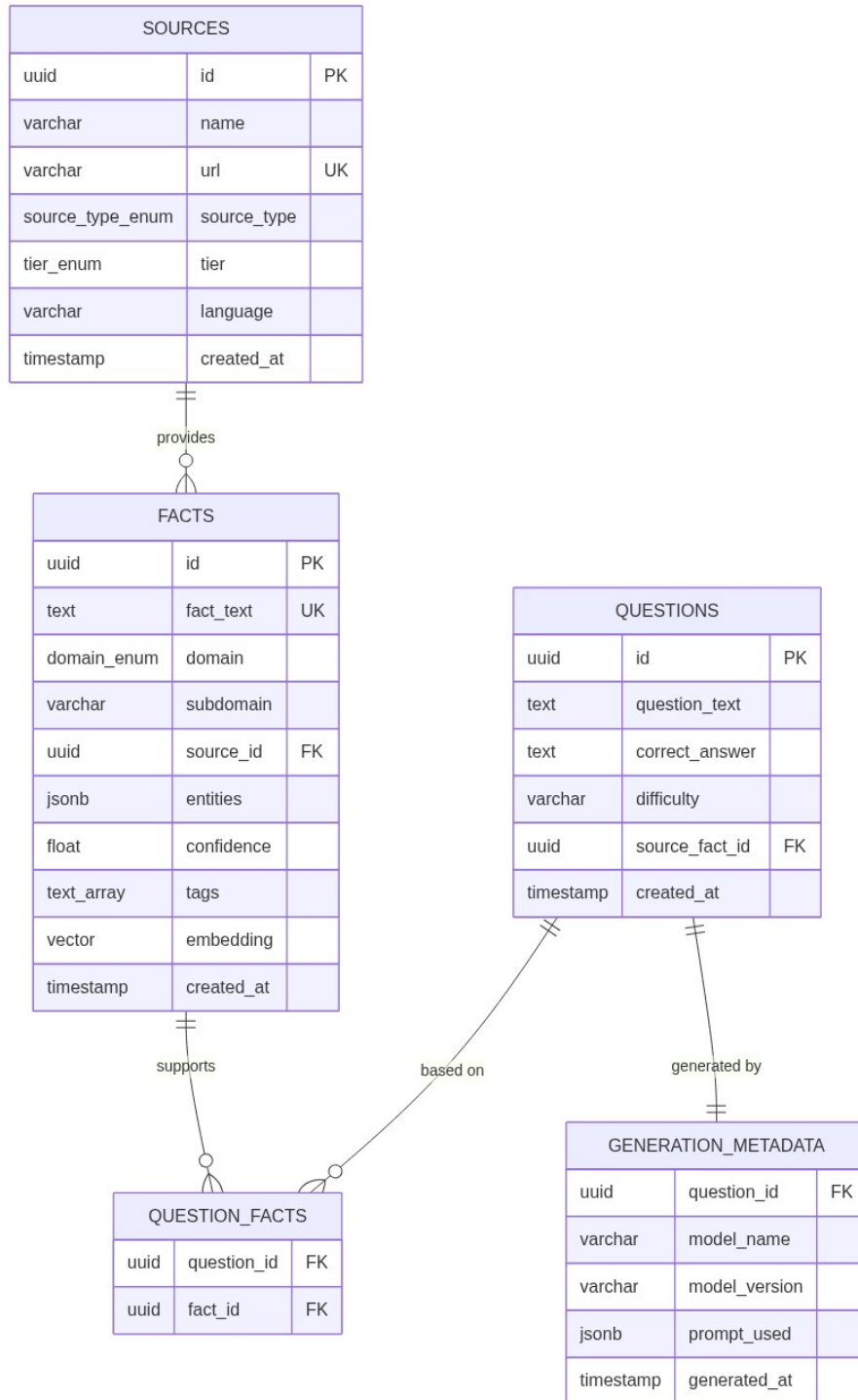


Figure 19: Simplified release schema. sources ground facts; questions pull from facts via question_facts; generation_metadata records the LLM and strategy that produced each question; audit_findings holds per-agent verdicts; human_reviews holds gold-sheet ratings.